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Deep Learning

Alberto Ezpondaburu

Analytical Index

1. Introduction to Deep Learning
2. Convolutional Neural Networks (CNN)
3. Recurrent Neural Networks (RNN)
4. Natural Language Processing with Deep Learning
5. Deep Generative Modelling

Analytical Index

1. Introduction to Deep Learning
2. Convolutional Neural Networks (CNN)
3. Recurrent Neural Networks (RNN)
4. Natural Language Processing with Deep Learning
5. Deep Generative Modelling
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    b. Autoencoders
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Deep Generative Models

Introduction

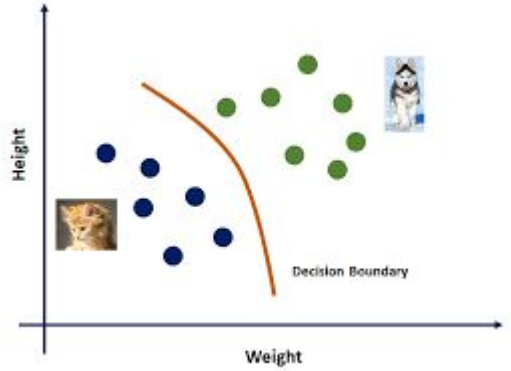


5.1

Generative vs Discriminative

Generative vs Discriminative models

Discriminative

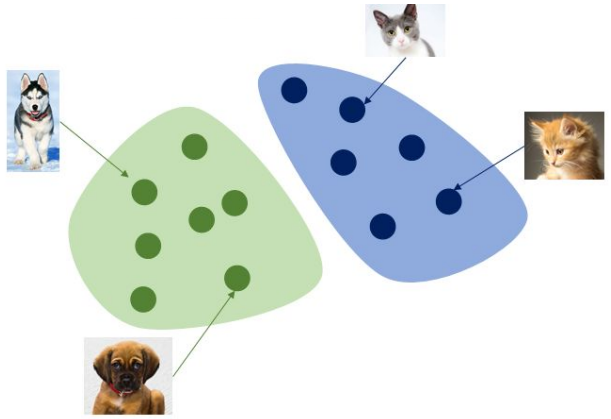


Model of the conditional probability of the target Y, given an observation x

Features $X \rightarrow Y$ Target

$$P(Y|X)$$

Generative



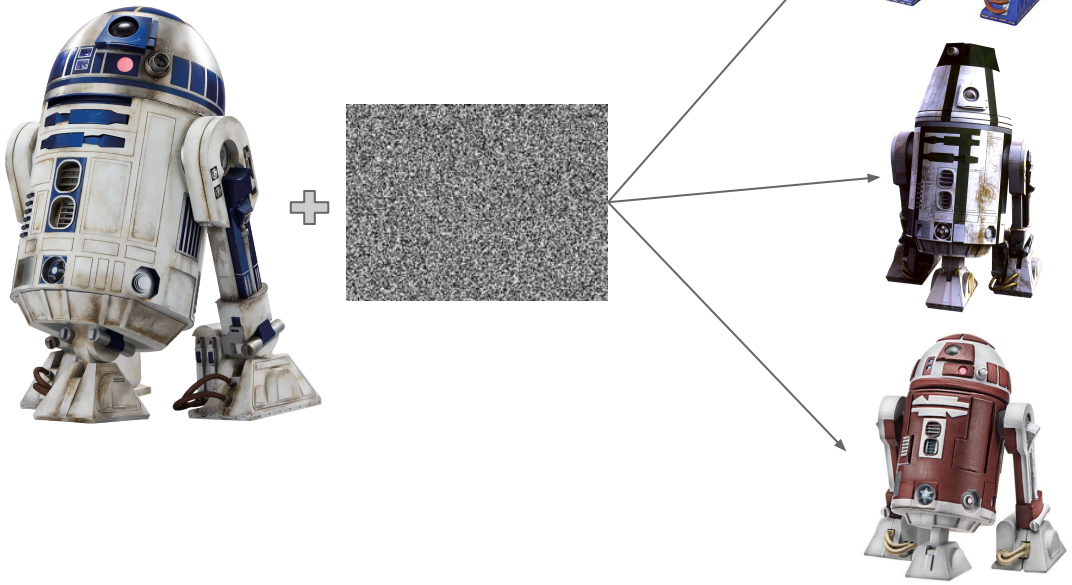
Target $Y \rightarrow X$ Features

$$P(X|Y)$$

Generative

Noise, Class Features

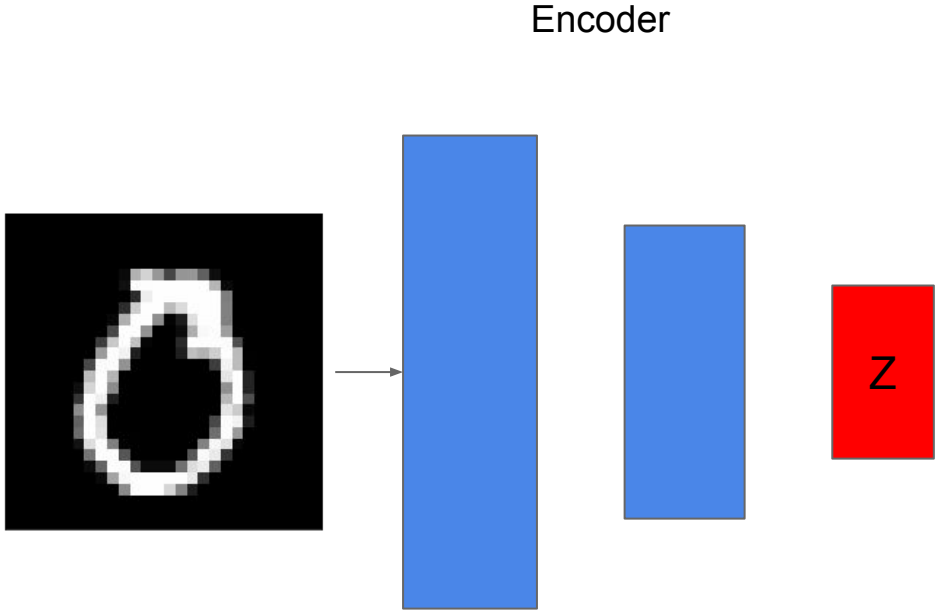
$$\xi, Y \rightarrow X$$



5.2

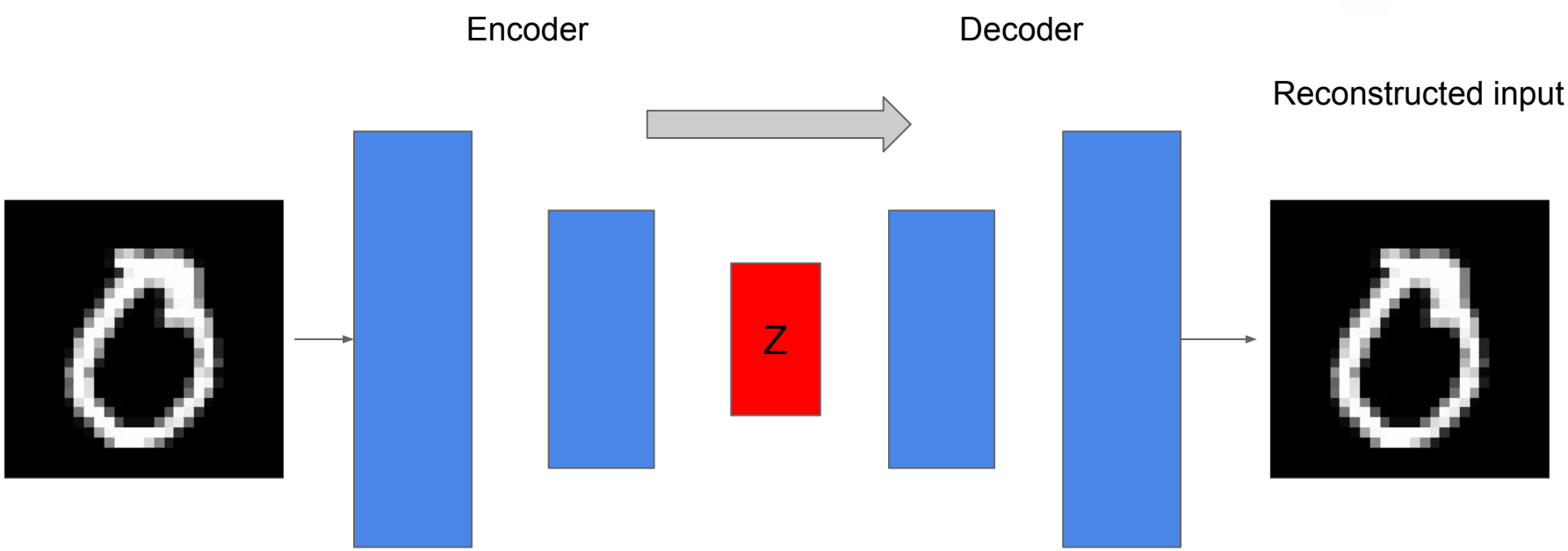
Autoencoders

Autoencoder Intuition



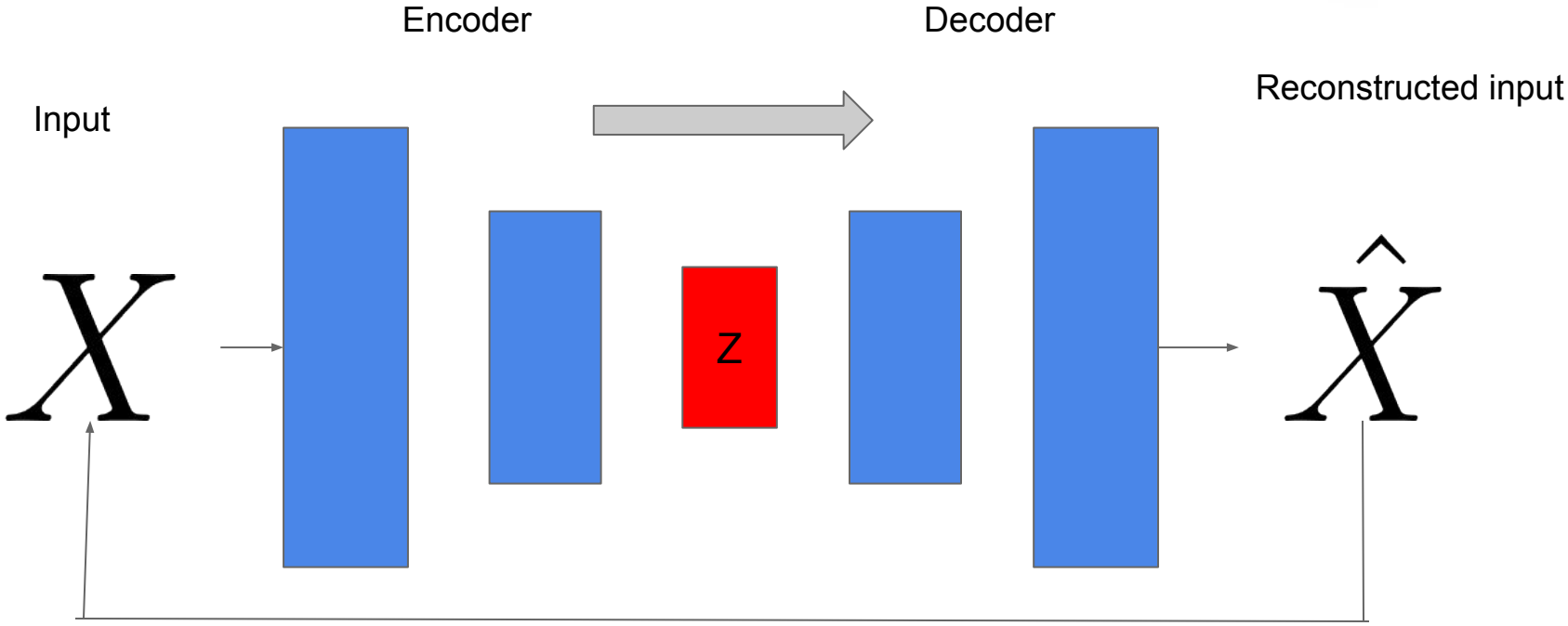
- We want to obtain a lower dimensional representation from the input data.
- Useful for dimensionality reduction and visualization.
- The encoder maps input data into a lower dimensional latent space (z)

Autoencoder Intuition



Train the model to reconstruct the original input data.
The decoder learns to reconstruct the data with the latent space.

Autoencoder: Reconstruction Loss



$$\mathcal{L}(X, \hat{X}, W) = ||X - \hat{X}||^2$$

Autoencoder Applications: Compression

2D latent space



5D latent space

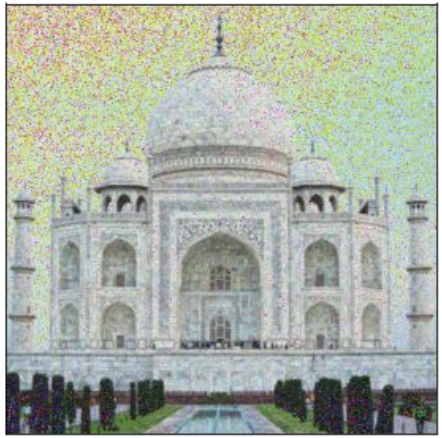


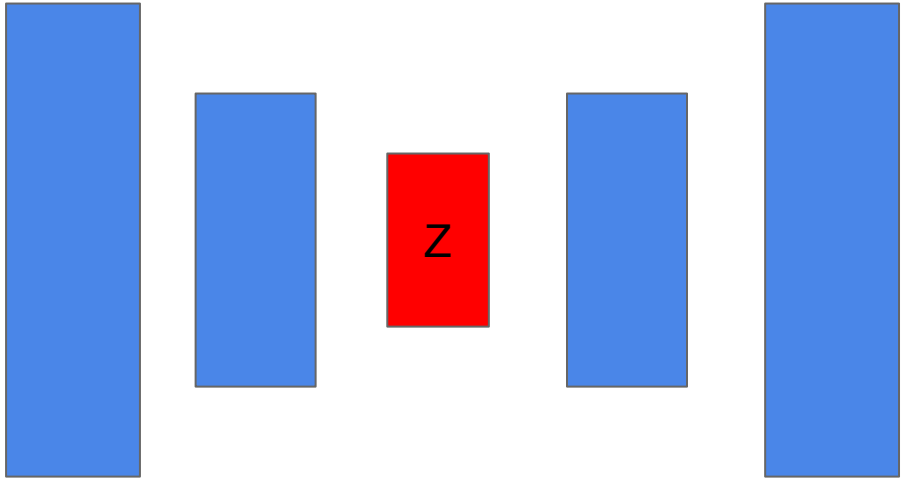
Ground Truth



Autoencoder Applications: Denoising

Noisy Image

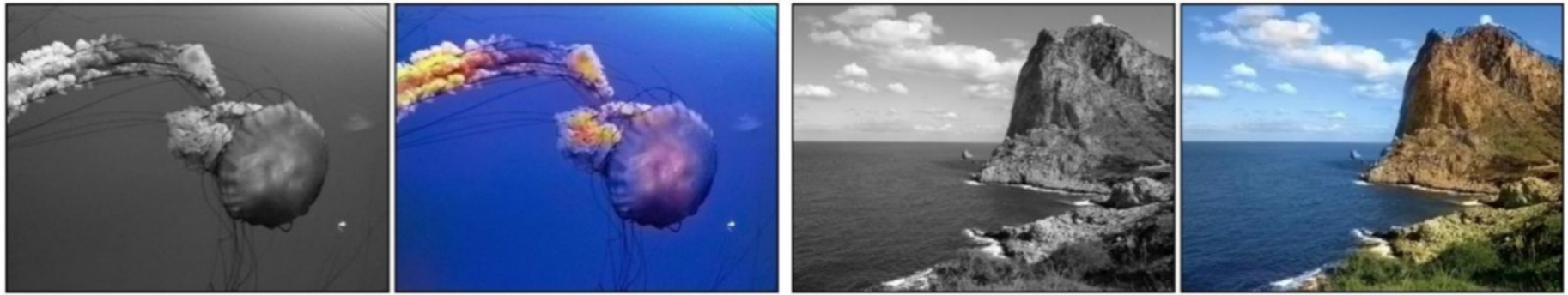




Denoised Image



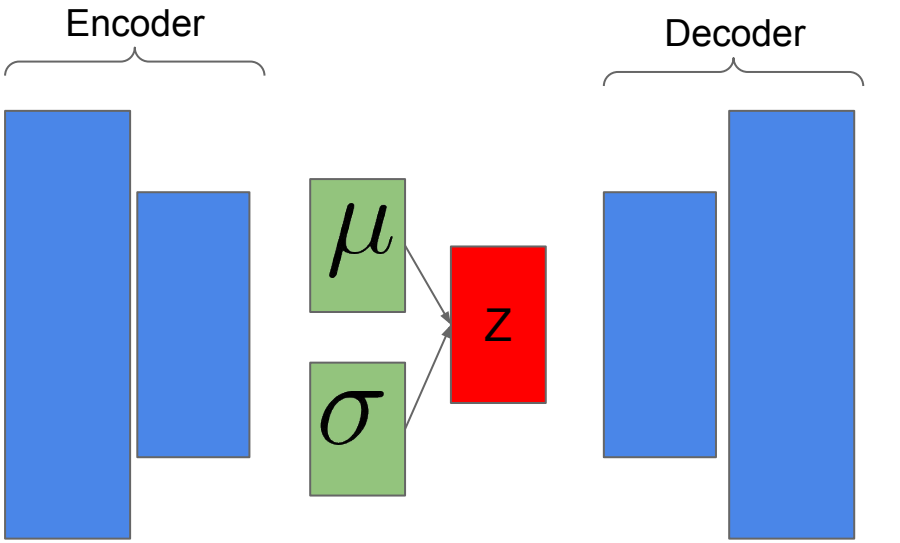
Autoencoder Applications: Colorizing Images



5.3

Variational Autoencoders

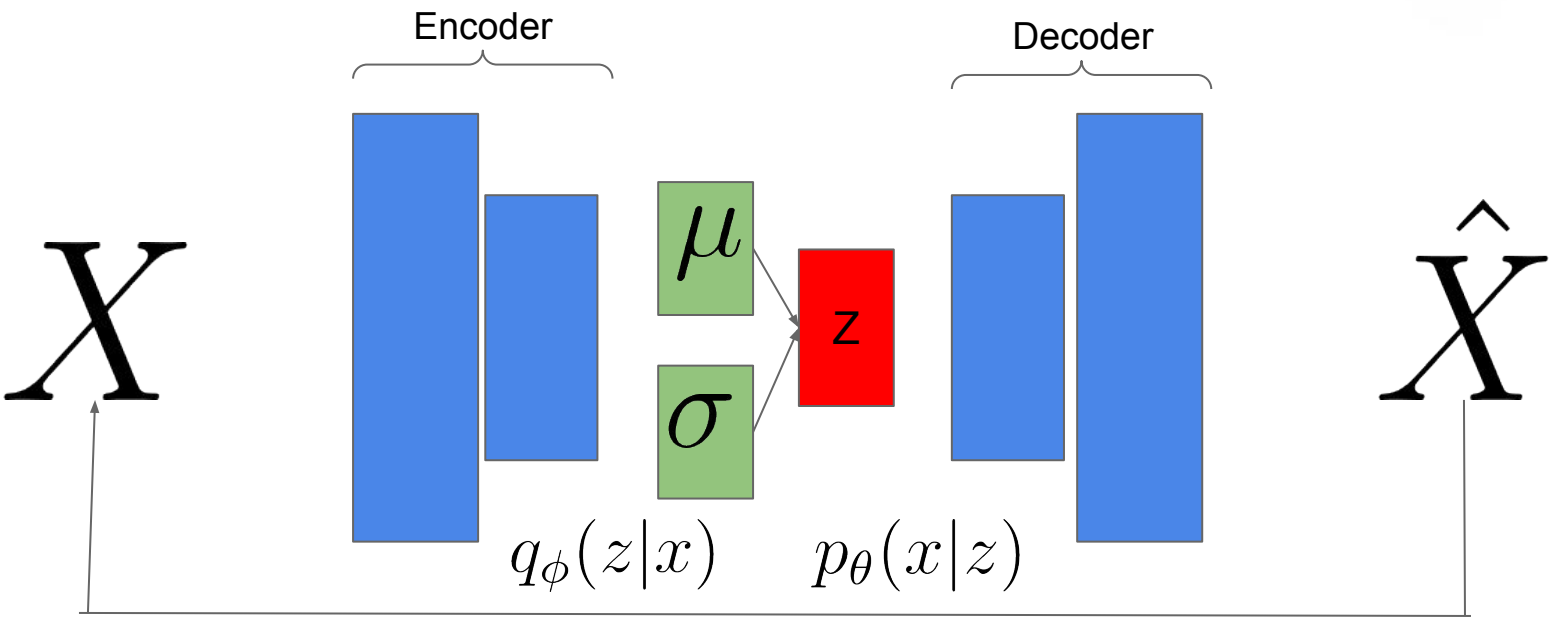
VAEs



Encoder computes: $q_\phi(z|x)$

Decoder computes: $p_\theta(x|z)$

VAEs: Loss



$$\mathcal{L} = \underbrace{||X - \hat{X}||^2}_{\text{Reconstruction loss}} + \underbrace{D(q_\phi(z|x)||p(z))}_{\text{Regularization}}$$

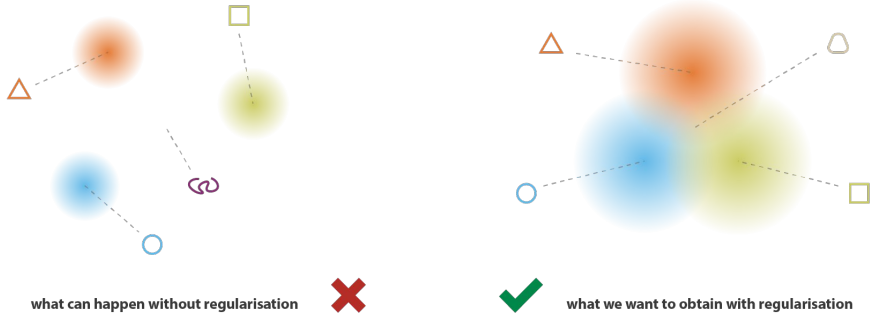
VAEs: Regularization

$$D(q_\phi(z|x)||p(z))$$

Inferred Distribution

$$\mathcal{N}(0,\ 1)$$

Continuity (two close points in the latent space should not give two completely different contents once decoded)
Completeness (for a chosen distribution, a point sampled from the latent space should give “meaningful” content once decoded).



https://towardsdatascience.com/understanding-variational-autoencoders-vaes-f70510919f73

VAEs



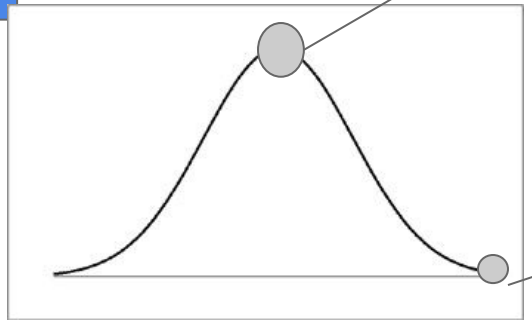
Encoder

Decoder

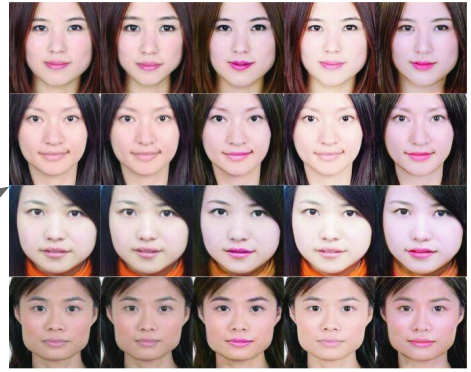
μ

σ

z



Homogeneous color, pose ...



Diverse



Music Generation



Music Generation



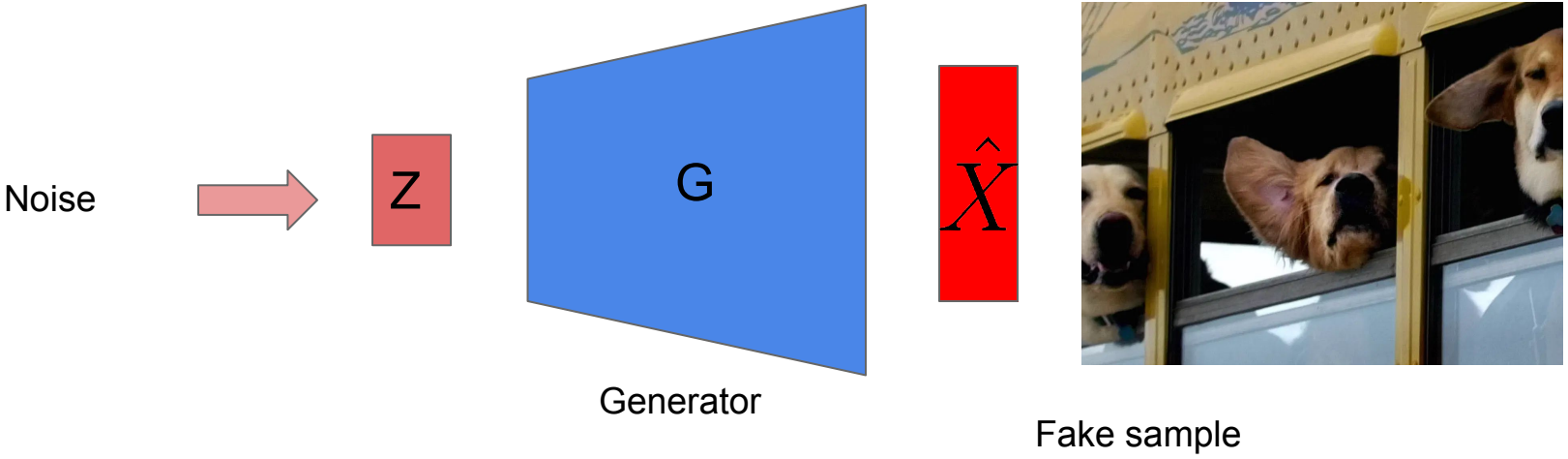
https://magenta.tensorflow.org/magenta-realtime

5.4

GANs: Generative Adversarial Networks

GANs: Why do we need to learn the distribution?

Instead of learning the density of the data, why don't we train a model that samples it directly?



GANs: Generator & Discriminator Intuition

Generative ADVERSARIAL Network: We have two competing neural networks.

Generator

Learn to sample **fake** data, that **looks real**.

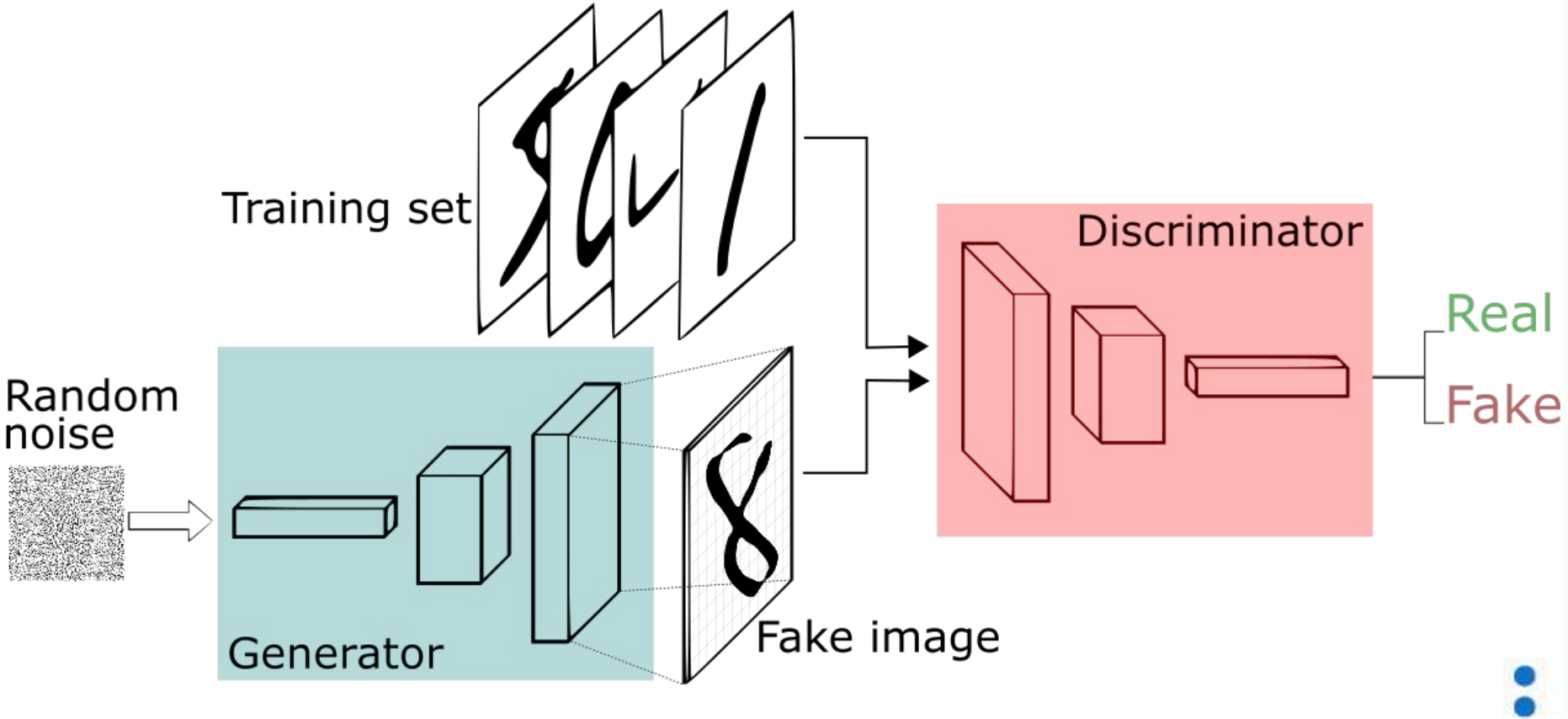


Discriminator

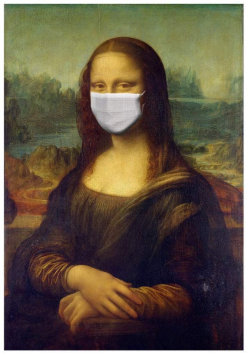
Learns to **distinguish** real from fake data.



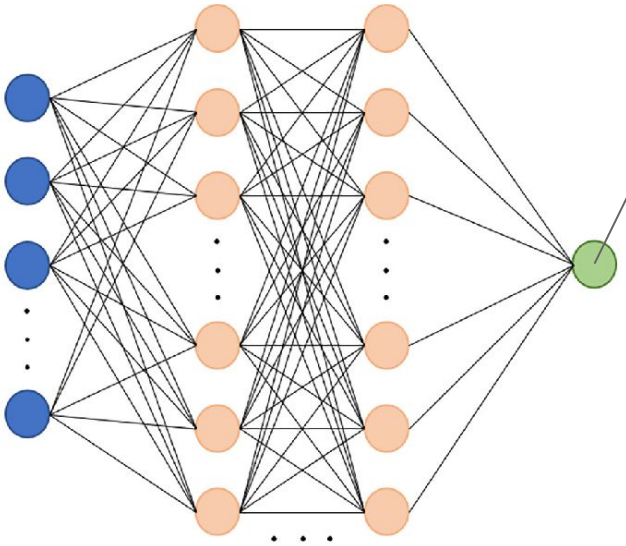
GANs: Generator & Discriminator



GANs: Discriminator



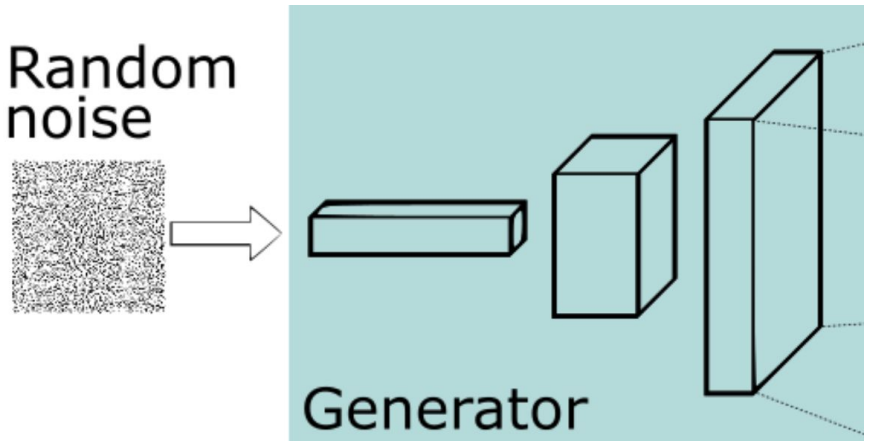


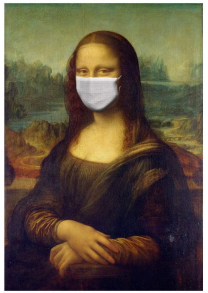


Fake or Real ??

- Is a binary classifier.
- Learns the probability of class Y (**fake** or **real**) given input X.
- Probabilities are the feedback for the generator.

GANs: Generator



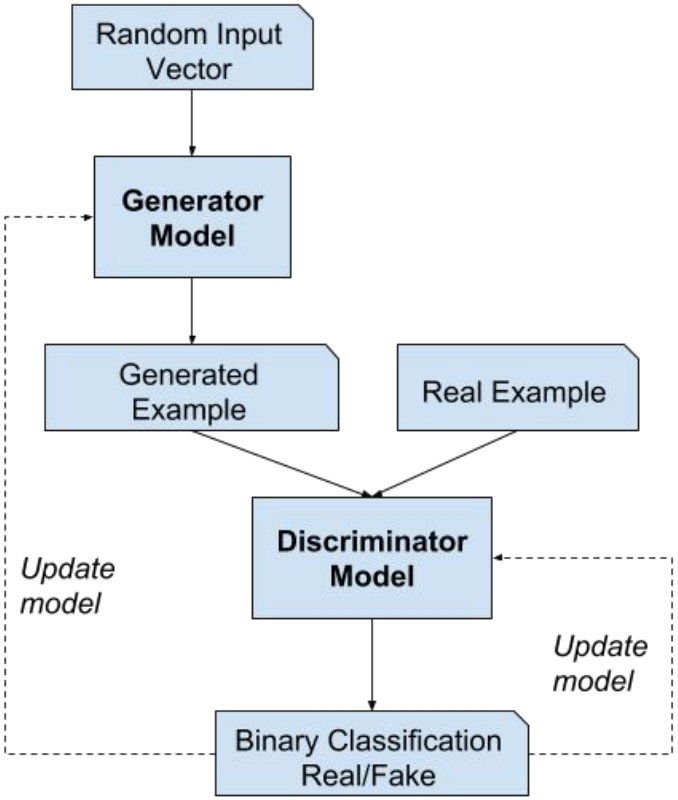


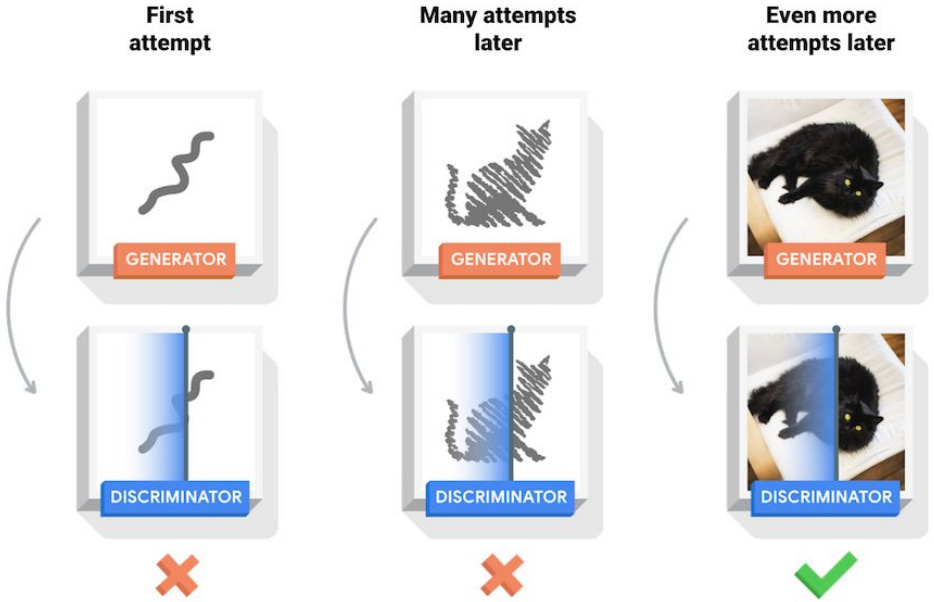




- Produces fake data.
- Learns the approximate distribution of inputs data.
- Takes noise as input

GANs: Training





Generator and discriminator, are trained together. The generator generates a batch of samples, and combined with real data are provided to the discriminator and classified as real or fake.

GANs: StyleGAN

StyleGAN is a novel generative adversarial network (GAN) introduced by Nvidia researchers in December 2018



GANs: StyleGAN



GANs: StyleGAN 2



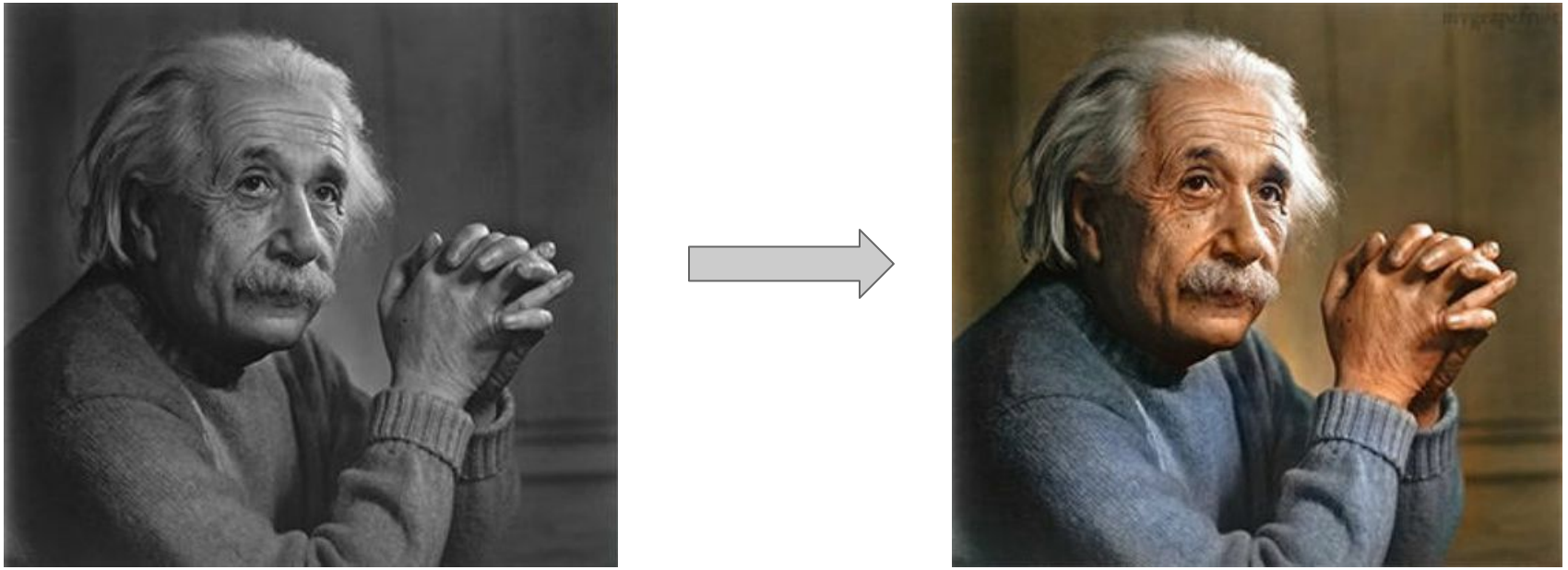
GANs: Create Anime Characters



Towards the Automatic Anime Characters Creation with Generative Adversarial Networks: https://arxiv.org/abs/1708.05509

GANs: Image to Image Translation

Image-to-image translation task of taking images from one domain and transforming them so they have the style (or characteristics) of images from another domain.



GANs: Pix2Pix Image to Image Translation

Paired translation: Train with pairs of images of the two domains

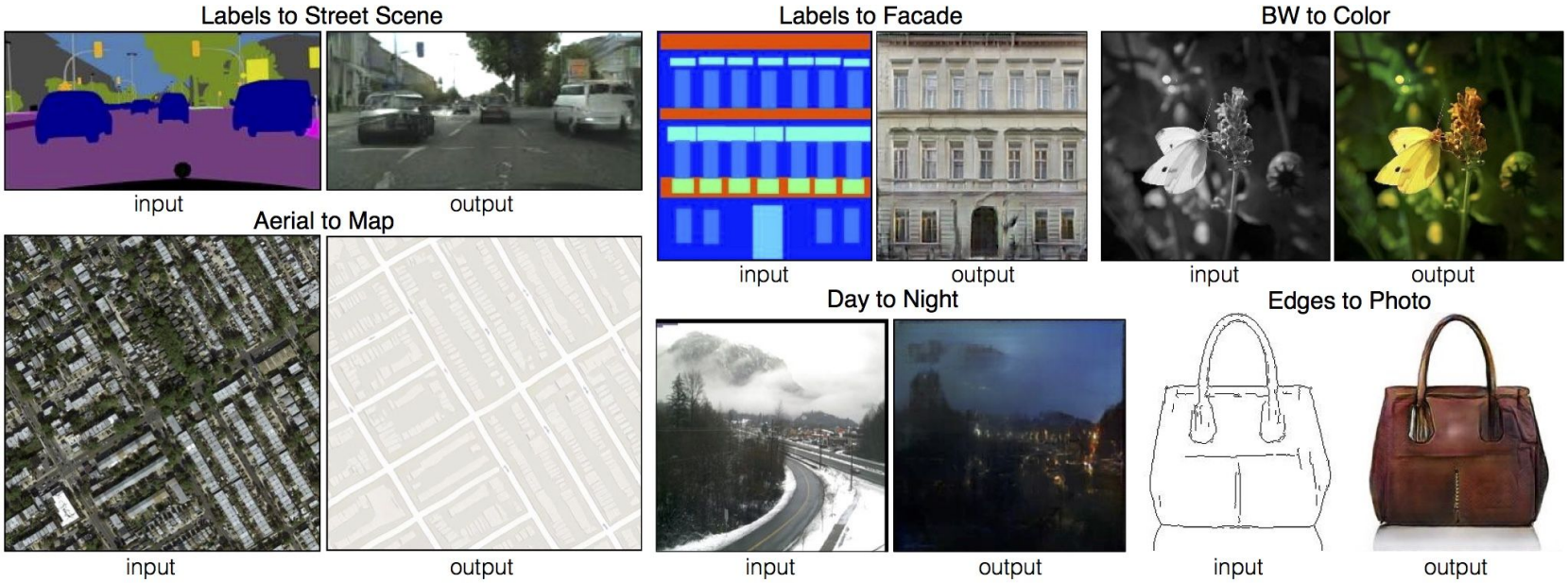
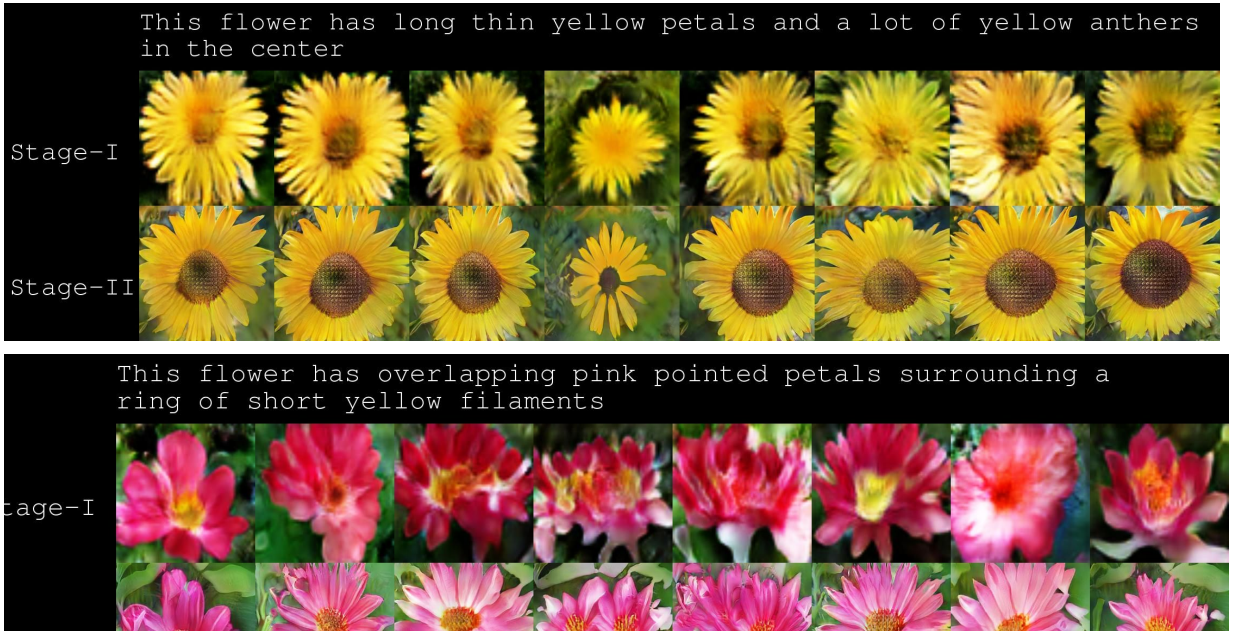


Image-to-Image Translation with Conditional Adversarial Networks

GANs: Face Aging



GANs: Text to Image StackGAN



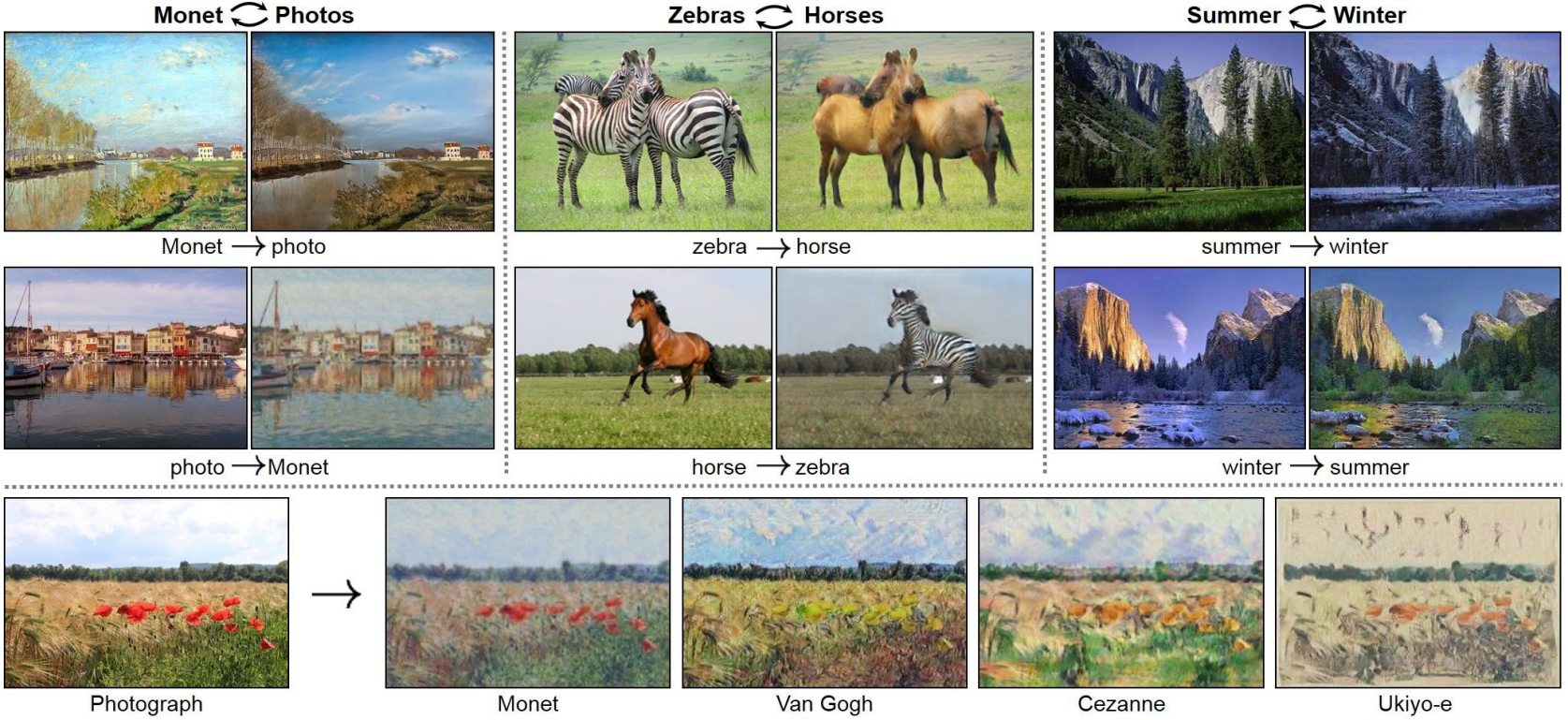
https://github.com/hanzhanggit/StackGAN

GANs: CycleGAN, Image to Image Translation

Unpaired image-to-image: Capture the characteristics of one image domain and figure out how these characteristics could be translated into another image domain.



GANs: CycleGAN, Image to Image Translation



GANs: Multimodal MUNIT



(a) edges ↔ shoes



(b) edges ↔ handbags

https://github.com/NVlabs/MUNIT

GANs: GauGAN 2



GANs



Few-Shot Adversarial Learning of Realistic Neural Talking Head Models

GANs: DeepFake



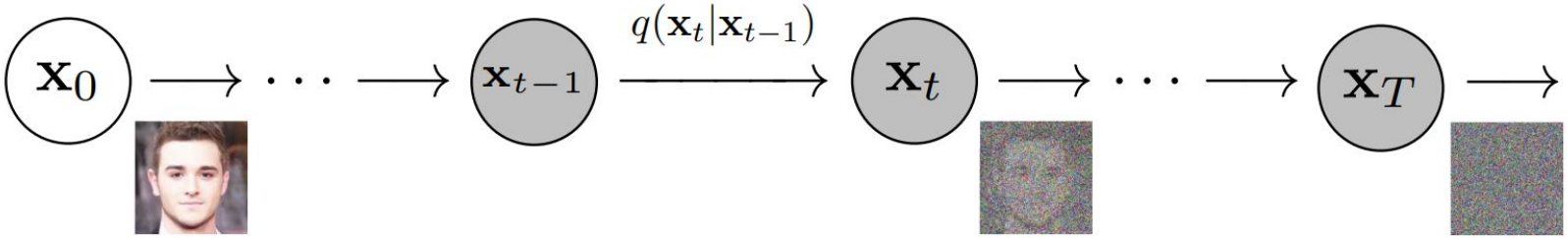
5.4

Diffusion Models and Other Projects

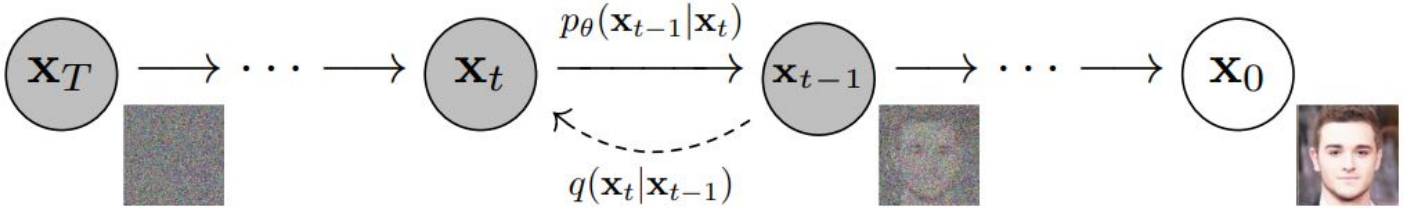
Diffusion models

The Diffusion Process

- Diffusion process: original data sample is progressively corrupted by adding noise.



- Reverse process (denoising): learn to recover the data by reversing this noising process.



[Denoising Diffusion Probabilistic Models Paper]

Diffusion models: DALL-E 2



[Denoising Diffusion Probabilistic Models Paper]

Diffusion models: DALL-E 3



Denoising Diffusion Probabilistic Models Paper

MusicLM: Generating Music From Text

Audio Generation From Rich Captions

Caption	Generated audio
The main soundtrack of an arcade game. It is fast-paced and upbeat, with a catchy electric guitar riff. The music is repetitive and easy to remember, but with unexpected sounds, like cymbal crashes or drum rolls.	0:00 / 0:30
A fusion of reggaeton and electronic dance music, with a spacey, otherworldly sound. Induces the experience of being lost in space, and the music would be designed to evoke a sense of wonder and awe, while being danceable.	0:00 / 0:30
A rising synth is playing an arpeggio with a lot of reverb. It is backed by pads, sub bass line and soft drums. This song is full of synth sounds creating a soothing and adventurous atmosphere. It may be playing at a festival during two songs for a buildup.	0:00 / 0:30
Slow tempo, bass-and-drums-led reggae song. Sustained electric guitar. High-pitched bongos with ringing tones. Vocals are relaxed with a laid-back feel, very expressive.	0:00 / 0:30

https://google-research.github.io/seanet/musiclm/examples/

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